

3.0.47

RAVULA SHASHANK REDDY - EE25BTECH11047

Problem

A memoryless source emits n symbols each with a probability p . The entropy of the source as a function of n is:

- (a) increases as $\log n$
- (b) decreases as $\log(1/n)$
- (c) increases as n
- (d) increases as $n \log n$

Solution

Since all n symbols are equally likely,

$$p = \frac{1}{n} \quad (1)$$

The entropy of a discrete memoryless source is:

$$H = - \sum_{i=1}^n p \log_2 p \quad (2)$$

Substituting $p = \frac{1}{n}$:

$$H = - \sum_{i=1}^n \frac{1}{n} \log_2 \left(\frac{1}{n} \right) \quad (3)$$

$$= -n \cdot \frac{1}{n} \log_2 \left(\frac{1}{n} \right) \quad (4)$$

$$= -\log_2 \left(\frac{1}{n} \right) \quad (5)$$

$$= \log_2 n \quad (6)$$

Final Answer:

$$H = \log_2 n \quad (7)$$

Option (a): increases as $\log n$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(1, 20)
H = np.log2(n)

plt.plot(n, H)
plt.xlabel('n')
plt.ylabel('Entropy H (bits)')
plt.grid()
plt.show()
```

Plot

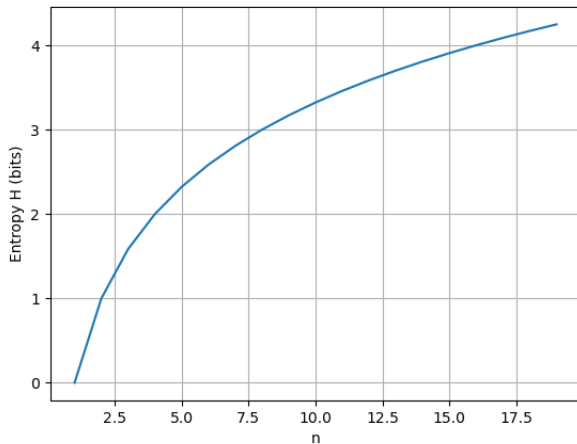


Figure: